MAST10005 Assignment 2

Semester 2 2023

Due Date: Thursday 24 August

Assignments are due in Gradescope at 11:59pm on the due date listed above. Late assignments are not accepted. Please see the instructions on the page where you accessed this file for details on submitting your assignment in Gradescope.

Students are encouraged to work together on understanding the problems and their solutions. However, submitted solutions must be prepared individually, in your own words, and without the aid of others. Answers presented without justification will receive no marks.

If you do not completely understand your submitted solutions, it is possible that you are committing academic misconduct. This is a serious offense.

Do not upload this assignment paper or any of its questions/solutions to any online *"help"* service (e.g., CourseHero, Chegg, etc...). By doing so, you are actively ruining the learning experience for yourself and future students. In doing so you are more than likely violating the University's academic misconduct policy.

Instructions: Use the boxes provided to write up your final draft of the work you are submitting to be assessed. You will need to scan these pages in order to submit to Gradescope.

You do not need to scan this cover page.

PART A – WebWork *(4 marks)*

Complete Part A using WebWork. See *Assignments* on Canvas for an Assignment named *Assignment 2 Webwork part* to complete this question.

There are two WebWork questions. You will need your result from Problem 2 in WebWork to complete Question 1 in Part B.

See over for PART B

1. In this question, we will use techniques involving complex numbers to study the sextic polynomial defined by

$$P(z) = z^6 - 4z^3 + 8.$$

(a) (1 mark) Use the quadratic formula to find the set of possible values of z^3 such that $P(z) = 0$. *(Note that we are not trying to find the values of z yet - only the values of z^3.)*

Writing $P(z) = (z^3)^2 - 4z^3 + 8$ and applying the quadratic formula gives

$$z^3 = \frac{4 \pm \sqrt{(-4)^2 - 4 \cdot 8}}{2} = \frac{4 \pm \sqrt{-4^2}}{2} = 2 \pm 2i.$$

(b) (2 marks) In the WebWork part of this assignment, you found the three solutions of $z^3 = 2 - 2i$ in exponential polar form.

Use a similar method to find the three solutions of $z^3 = 2 + 2i$. Express your answer in exponential polar form.

We have $2 + 2i = 2\sqrt{2}e^{i\frac{\pi}{4}}$.

Applying the root finding formula gives $z = (2\sqrt{2})^{1/3}e^{i\frac{\pi/4+2k\pi}{3}}$ for $k = 0, 1, 2$.

Thus the three solutions are

$$\sqrt{2}e^{i\frac{\pi}{12}}, \sqrt{2}e^{i\frac{3\pi}{4}}, \sqrt{2}e^{i\frac{17\pi}{12}} = \sqrt{2}e^{-i\frac{7\pi}{12}}.$$

(c) (1 mark) Use the results of the previous parts to find the six values of z such that $P(z) = 0$. Express your answers in exponential polar form.

By part (a), the solutions to $P(z) = 0$ are the three values found in the WebWork part of the assignment together with the three values found in part (b).

Thus

$$\{z \in \mathbb{C} \mid P(z) = 0\} = \left\{ \sqrt{2}e^{i\frac{\pi}{12}}, \sqrt{2}e^{i\frac{3\pi}{4}}, \sqrt{2}e^{-i\frac{7\pi}{12}}, \sqrt{2}e^{-i\frac{\pi}{12}}, \sqrt{2}e^{-i\frac{3\pi}{4}}, \sqrt{2}e^{i\frac{7\pi}{12}} \right\}.$$

(d) (3 marks) Use your answers from part (c) to factor the polynomial $P(z)$ into three quadratics with *real* coefficients. (*You may use the facts that* $\cos(\frac{\pi}{12}) = \frac{\sqrt{6}+\sqrt{2}}{4}$ *and* $\sin(\frac{\pi}{12}) = \frac{\sqrt{6}-\sqrt{2}}{4}$).

By part (c) we can write P as:

$$P(z) = (z - \sqrt{2}e^{i\frac{\pi}{12}})(z - \sqrt{2}e^{i\frac{3\pi}{4}})(z - \sqrt{2}e^{-i\frac{7\pi}{12}})$$
$$(z - \sqrt{2}e^{-i\frac{\pi}{12}})(z - \sqrt{2}e^{-i\frac{3\pi}{4}})(z - \sqrt{2}e^{i\frac{7\pi}{12}}).$$

Taking the product of factors involving conjugate pairs yields

$$\begin{aligned}
(z - \sqrt{2}e^{i\frac{\pi}{12}})(z - \sqrt{2}e^{-i\frac{\pi}{12}}) &= z^2 - 2\sqrt{2}\cos(\tfrac{\pi}{12})\, z + 2 \\
&= z^2 - (\sqrt{3}+1)z + 2 \\
(z - \sqrt{2}e^{i\frac{3\pi}{4}})(z - \sqrt{2}e^{-i\frac{3\pi}{4}}) &= z^2 - 2\sqrt{2}\cos(\tfrac{3\pi}{4})\, z + 2 = z^2 + 2z + 2 \\
(z - \sqrt{2}e^{i\frac{7\pi}{12}})(z - \sqrt{2}e^{-i\frac{7\pi}{12}}) &= z^2 - 2\sqrt{2}\cos(\tfrac{7\pi}{12})\, z + 2 \\
= z^2 + 2\sqrt{2}\sin(\tfrac{\pi}{12})\, z + 2 &= z^2 + (\sqrt{3}-1)z + 2.
\end{aligned}$$

Thus

$$P(z) = (z^2 - (\sqrt{3}+1)z + 2)(z^2 + (\sqrt{3}-1)z + 2)(z^2 + 2z + 2).$$

Before you submit this work, make sure to check your answer. If you multiply out your quadratic factors, you should recover $P(z)$.

2. (4 marks) Sketch the following subset of $\mathbb{C}$

$$A = \{z \in \mathbb{C} : |z^2| < 9\} \cap \{z \in \mathbb{C} : \mathrm{Arg}(z^3) \in \left(\tfrac{\pi}{2}, \pi\right)\}.$$

(*Explain your reasoning.*)

Since $|z^2| = |z|^2$ and $|z| \geq 0$, the condition $|z^2| < 9$ is equivalent to $|z| < 3$.

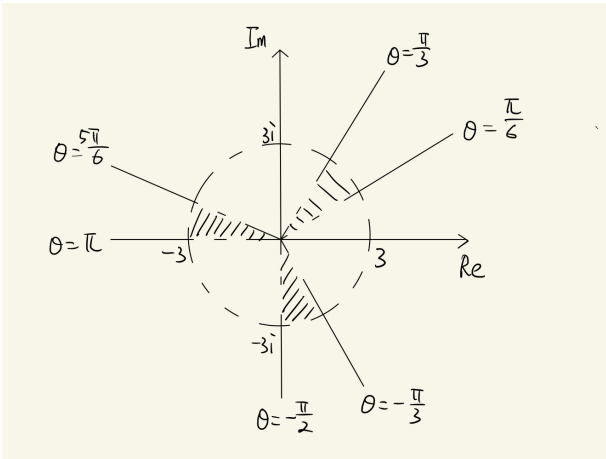
Suppose that $z = re^{i\theta}$, where $\theta = \mathrm{Arg}(z) \in (-\pi, \pi]$. Then $z^3 = r^3 e^{i3\theta}$.

Since 3θ is an argument of z^3 and lies in $(-3\pi, 3\pi]$, the condition that $\mathrm{Arg}(z^3) \in (\frac{\pi}{2}, \pi)$ becomes

$$\begin{aligned} 3\theta &\in (\tfrac{\pi}{2} - 2\pi, \pi - 2\pi) \cup (\tfrac{\pi}{2}, \pi) \cup (\tfrac{\pi}{2} + 2\pi, \pi + 2\pi) \\ &= (-\tfrac{3\pi}{2}, -\pi) \cup (\tfrac{\pi}{2}, \pi) \cup (\tfrac{5\pi}{2}, 3\pi) \end{aligned}$$

(this is the intersection of $(-3\pi, 3\pi]$ with the shifts of $(\frac{\pi}{2}, \pi)$ by multiples of 2π).

This becomes

$$\theta = \mathrm{Arg}(z) \in (-\tfrac{\pi}{2}, -\tfrac{\pi}{3}) \cup (\tfrac{\pi}{6}, \tfrac{\pi}{3}) \cup (\tfrac{5\pi}{6}, \pi).$$

Therefore the set A is as follows, where the boundaries are not inlcuded